

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change • No single source could meet demand all year; battery storage on the island grid smooths fluctuations • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated)	3
01.2	<i>Any 3 from:</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection	3
01.3	• Small groups can organise, agree and act quickly; decisions benefit people who see the results directly • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away	2
01.4	• Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	

Question	Marking guidance	Marks
02.1	- Coordinating global data on biodiversity conservation (Red List) - Increasing understanding of the importance of biodiversity - Deploying nature-based solutions to global challenges in climate, food and sustainable development	3
02.2	- Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered	3
02.3	- Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations	3
02.4	Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
03.1	- Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges - High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) - Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation - Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times	4
03.2	- Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range - Many species are rare with small ranges / small populations	2
03.3	- No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medically active in controlled doses (quinine, diosgenin, curare) - Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril)	2
03.4	- Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall - Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
04.1	• Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds • Water-repelling surface of lotus flowers • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to • Hooked burrs on seeds that stick to animal fur	4
04.2	• The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) • Air blowing over the mound has a lower pressure which also draws air out • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used	3
04.3	Any 2 from: • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs	2
04.4	• They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
05.1	• Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion	3
05.2	• Control of movement (housing / restricted grazing) so less energy is used in muscle activity • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage	3
05.3	• Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
06.1	<i>Any 2 from:</i> • Hydrological cycle • Carbon cycle • Nitrogen cycle • Atmospheric circulation • Thermohaline circulation • Photosynthesis	2
06.2	• Enzymes reduce the activation energy of reactions • So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures	2
06.3	<i>Any 1 from:</i> • Haber process (nitrogen fixation) • Incineration of wastes (decomposition) • Many manufacturing processes	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
07.1	• The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) • X survives from about pH 5.5 to 8.5 (wide range); Y only from about pH 6.0 to 8.0 (narrow range) • Both have the same optimum (pH 7.0)	3
07.2	• X: 60% survive at pH 6.0 whereas only 5% of Y survive • X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions • Survivors breed and their offspring are adapted so the population rebuilds; Y is likely to die out locally	3
07.3	• Reveals the habitat requirements / which abiotic factor is near the edge of tolerance • Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival	2
07.4	<i>Any 2 from:</i> • Food supply • Pollinators • Seed dispersal species • Disease / pathogens • Predators / competitors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	
08.1	• Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump • About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages • Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice • Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth	4

Question	Marking guidance	Marks
08.2	• Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour • Fewer young trees eaten so trees and riverside vegetation recovered • More trees allowed beaver populations to increase; beaver dams increased wetland area • Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers	4
08.3	• Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species • Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	
09.1	• In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate	4
09.2	• Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor	2
09.3	• Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy)	3
09.4	• Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
10.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25